

Women Safety Band – Mobile App Prototype

1. Introduction

The *Women Safety Band System* is a smart wearable and mobile application ecosystem designed to provide **real-time emergency assistance, live tracking, and evidence capture** for women in critical situations.

This system is built to function **independently**, ensuring safety even when the user's mobile phone is unavailable.

2. System Overview

The system consists of two primary components:

A. Smart Wearable Band

- Independent emergency trigger device
- Sends real-time alerts, GPS location, audio/video data

B. Mobile Application

- Monitoring and control interface
 - Displays live incident data and manages alerts
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3. Connectivity Architecture

3.1 Core Principle

The device operates as a **fully independent, long-range connected system**, not dependent on short-range technologies.

3.2 Technologies Used

Technology	Purpose
LTE-M / NB-IoT (Cellular)	Long-range communication
GPS Module	Real-time location tracking
Cloud Server	Data processing & alert routing
eSIM / SIM Connectivity	Always-on network access

3.3 Working Model

Normal Mode

- Device remains connected to cloud server
 - Periodically updates status (battery, connectivity)
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Emergency Mode (Core Functionality)

When the panic trigger is activated:

1. Band directly transmits data via cellular network
2. Sends:
 - Live GPS location
 - Audio/video stream
 - Emergency alert signal
3. Cloud server processes the alert and:
 - Notifies trusted contacts
 - Alerts police/emergency services
4. Mobile application:
 - Receives alert instantly
 - Displays live tracking and media feed

Ensures functionality regardless of phone availability.

3.4 System Flow

Wearable Band



(Cellular Network: LTE-M / NB-IoT)



Cloud Server



Mobile App + Contacts + Police

4. Mobile Application – Feature Overview

4.1 Onboarding

- User registration & authentication
 - Device ID linking (manual or QR-based)
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4.2 Home Screen

- Large emergency panic button
 - Device battery status
 - Network connectivity status
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4.3 Emergency Activation

Upon trigger:

- Live GPS tracking starts
 - Audio/video streaming begins
 - Alerts sent instantly to:
 - Contacts
 - Authorities
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4.4 Live Incident Screen

- Real-time map tracking
 - Live video feed
 - Audio streaming indicator
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4.5 Trusted Contacts

- Add/remove contacts
 - Configure alert methods:
 - SMS
 - Call
 - App notification
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4.6 Device Status

- Battery percentage
 - Network strength
 - Device health status
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4.7 Incident History

- Stores past emergency events
- Includes:
 - Time & date
 - Location
 - Media evidence

Enables strong legal proof support.

5. Prototype Development Plan

5.1 Objective

To develop a **functional mobile app prototype** that demonstrates:

- UI/UX design
 - Emergency workflow
 - Real-time simulation
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5.2 Tools & Technologies

Component	Tool
UI Design	Figma
App Development	Flutter / React Native
Backend	Firebase
Maps	Google Maps API
Notifications	Firebase / SMS API

5.3 Development Steps

Step 1: UI Design

Design core screens:

- Onboarding
 - Home
 - Live Incident
 - Contacts
 - History
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Step 2: Device Simulation

- Simulate wearable trigger using in-app button
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Step 3: Backend Setup

- Configure Firebase
 - Set up database & authentication
 - Implement alert logic
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Step 4: Feature Simulation

- GPS via mobile device
 - Camera for live video simulation
 - Push notifications for alerts
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Step 5: Demo Workflow

- Trigger emergency
 - Display live tracking
 - Show alert system
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6. Connectivity Logic (Prototype)

- Fully cloud-based communication
- No dependency on short-range connectivity
- Fallback:
 - SMS-based location sharing in low network conditions

Aligns with real product architecture.

7. UI/UX Design Guidelines

- Primary color: Red (#D33) for emergency
 - Dark background for clarity
 - Minimal and fast-access interface
 - Large interactive elements
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8. Key Advantages

- Works independently of smartphone
 - Real-time emergency response
 - Evidence generation (audio/video/location)
 - Scalable for nationwide deployment
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9. Conclusion

The Women Safety Band system represents a **next-generation safety solution**, combining wearable technology with cloud-based intelligence to deliver:

- Instant response
- Reliable communication
- Actionable evidence

The prototype will validate the system's usability and demonstrate its readiness for real-world implementation.